

Additional Topics

1.A Graphs and Free Groups

Since all groups can be realized as fundamental groups of spaces, this opens the way for using topology to study algebraic properties of groups. The topics in this section and the next give some illustrations of this principle, mainly using covering space theory.

We remind the reader that the Additional Topics which form the remainder of this chapter are not to be regarded as an essential part of the basic core of the book. Readers who are eager to move on to new topics should feel free to skip ahead.

By definition, a **graph** is a 1-dimensional CW complex, in other words, a space X obtained from a discrete set X^0 by attaching a collection of 1-cells e_α . Thus X is obtained from the disjoint union of X^0 with closed intervals I_α by identifying the two endpoints of each I_α with points of X^0 . The points of X^0 are the **vertices** and the 1-cells the **edges** of X . Note that with this definition an edge does not include its endpoints, so an edge is an open subset of X . The two endpoints of an edge can be the same vertex, so the closure $\bar{e}_\alpha$ of an edge e_α is homeomorphic either to I or S^1 .

Since X has the quotient topology from the disjoint union $X^0 \coprod_\alpha I_\alpha$, a subset of X is open (or closed) iff it intersects the closure $\bar{e}_\alpha$ of each edge e_α in an open (or closed) set in $\bar{e}_\alpha$. One says that X has the **weak topology** with respect to the subspaces $\bar{e}_\alpha$. In this topology a sequence of points in the interiors of distinct edges forms a closed subset, hence never converges. This is true in particular if the edges containing the sequence all have a common vertex and one tries to choose the sequence so that it gets 'closer and closer' to the vertex. Thus if there is a vertex that is the endpoint of infinitely many edges, then the weak topology cannot be a metric topology. An exercise at the end of this section is to show the converse, that the weak topology is a metric topology if each vertex is an endpoint of only finitely many edges.

A basis for the topology of X consists of the open intervals in the edges together with the path-connected neighborhoods of the vertices. A neighborhood of the latter sort about a vertex v is the union of connected open neighborhoods U_α of v in $\bar{e}_\alpha$ for all $\bar{e}_\alpha$ containing v . In particular, we see that X is locally path-connected. Hence a graph is connected iff it is path-connected.

If X has only finitely many vertices and edges, then X is compact, being the continuous image of the compact space $X^0 \coprod_\alpha I_\alpha$. The converse is also true, and more generally, a compact subset C of a graph X can meet only finitely many vertices and edges of X . To see this, let the subspace $D \subset C$ consist of the vertices in C together with one point in each edge that C meets. Then D is a closed subset of X since it

meets each $\bar{e}_\alpha$ in a closed set. For the same reason, any subset of D is closed, so D has the discrete topology. But D is compact, being a closed subset of the compact space C , so D must be finite. By the definition of D this means that C can meet only finitely many vertices and edges.

A **subgraph** of a graph X is a subspace $Y \subset X$ that is a union of vertices and edges of X , such that $e_\alpha \subset Y$ implies $\bar{e}_\alpha \subset Y$. The latter condition just says that Y is a closed subspace of X . A **tree** is a contractible graph. By a tree in a graph X we mean a subgraph that is a tree. We call a tree in X **maximal** if it contains all the vertices of X . This is equivalent to the more obvious meaning of maximality, as we will see below.

Proposition 1A.1. *Every connected graph contains a maximal tree, and in fact any tree in the graph is contained in a maximal tree.*

Proof: Let X be a connected graph. We will describe a construction that embeds an arbitrary subgraph $X_0 \subset X$ as a deformation retract of a subgraph $Y \subset X$ that contains all the vertices of X . By choosing X_0 to be any subtree of X , for example a single vertex, this will prove the proposition.

As a preliminary step, we construct a sequence of subgraphs $X_0 \subset X_1 \subset X_2 \subset \cdots$, letting X_{i+1} be obtained from X_i by adjoining the closures $\bar{e}_\alpha$ of all edges $e_\alpha \subset X - X_i$ having at least one endpoint in X_i . The union $\bigcup_i X_i$ is open in X since a neighborhood of a point in X_i is contained in X_{i+1} . Furthermore, $\bigcup_i X_i$ is closed since it is a union of closed edges and X has the weak topology. So $X = \bigcup_i X_i$ since X is connected.

Now to construct Y we begin by setting $Y_0 = X_0$. Then inductively, assuming that $Y_i \subset X_i$ has been constructed so as to contain all the vertices of X_i , let Y_{i+1} be obtained from Y_i by adjoining one edge connecting each vertex of $X_{i+1} - X_i$ to Y_i , and let $Y = \bigcup_i Y_i$. It is evident that Y_{i+1} deformation retracts to Y_i , and we may obtain a deformation retraction of Y to $Y_0 = X_0$ by performing the deformation retraction of Y_{i+1} to Y_i during the time interval $[1/2^{i+1}, 1/2^i]$. Thus a point $x \in Y_{i+1} - Y_i$ is stationary until this interval, when it moves into Y_i and thereafter continues moving until it reaches Y_0 . The resulting homotopy $h_t : Y \rightarrow Y$ is continuous since it is continuous on the closure of each edge and Y has the weak topology. $\square$

Given a maximal tree $T \subset X$ and a base vertex $x_0 \in T$, then each edge e_α of $X - T$ determines a loop f_α in X that goes first from x_0 to one endpoint of e_α by a path in T , then across e_α , then back to x_0 by a path in T . Strictly speaking, we should first orient the edge e_α in order to specify which direction to cross it. Note that the homotopy class of f_α is independent of the choice of the paths in T since T is simply-connected.

Proposition 1A.2. *For a connected graph X with maximal tree T , $\pi_1(X)$ is a free group with basis the classes $[f_\alpha]$ corresponding to the edges e_α of $X - T$.*

In particular this implies that a maximal tree is maximal in the sense of not being contained in any larger tree, since adjoining any edge to a maximal tree produces a graph with nontrivial fundamental group. Another consequence is that a graph is a tree iff it is simply-connected.

Proof: The quotient map $X \rightarrow X/T$ is a homotopy equivalence by Proposition 0.17. The quotient X/T is a graph with only one vertex, hence is a wedge sum of circles, whose fundamental group we showed in Example 1.21 to be free with basis the loops given by the edges of X/T , which are the images of the loops f_α in X . $\square$

Here is a very useful fact about graphs:

Lemma 1A.3. *Every covering space of a graph is also a graph, with vertices and edges the lifts of the vertices and edges in the base graph.*

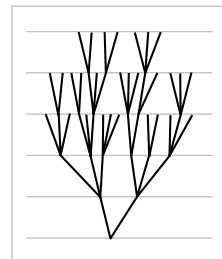
Proof: Let $p: \tilde{X} \rightarrow X$ be the covering space. For the vertices of $\tilde{X}$ we take the discrete set $\tilde{X}^0 = p^{-1}(X^0)$. Writing X as a quotient space of $X^0 \coprod_\alpha I_\alpha$ as in the definition of a graph and applying the path lifting property to the resulting maps $I_\alpha \rightarrow X$, we get a unique lift $I_\alpha \rightarrow \tilde{X}$ passing through each point in $p^{-1}(x)$, for $x \in e_\alpha$. These lifts define the edges of a graph structure on $\tilde{X}$. The resulting topology on $\tilde{X}$ is the same as its original topology since both topologies have the same basic open sets, the covering projection $\tilde{X} \rightarrow X$ being a local homeomorphism. $\square$

We can now apply what we have proved about graphs and their fundamental groups to prove a basic fact of group theory:

Theorem 1A.4. *Every subgroup of a free group is free.*

Proof: Given a free group F , choose a graph X with $\pi_1(X) \approx F$, for example a wedge of circles corresponding to a basis for F . For each subgroup G of F there is by Proposition 1.36 a covering space $p: \tilde{X} \rightarrow X$ with $p_*(\pi_1(\tilde{X})) = G$, hence $\pi_1(\tilde{X}) \approx G$ since p_* is injective by Proposition 1.31. Since $\tilde{X}$ is a graph by the preceding lemma, the group $G \approx \pi_1(\tilde{X})$ is free by Proposition 1A.2. $\square$

The structure of trees can be elucidated by looking more closely at the constructions in the proof of Proposition 1A.1. If X is a tree and v_0 is any vertex of X , then the construction of a maximal tree $Y \subset X$ starting with $Y_0 = \{v_0\}$ yields an increasing sequence of subtrees $Y_n \subset X$ whose union is all of X since a tree has only one maximal subtree, namely itself. We can think of the vertices in $Y_n - Y_{n-1}$ as being at ‘height’ n , with the edges of $Y_n - Y_{n-1}$ connecting these vertices to vertices of height $n - 1$. In this way we get a ‘height function’ $h: X \rightarrow \mathbb{R}$ assigning to each vertex its height, and monotone on edges.



For each vertex v of X there is exactly one edge leading downward from v , so by following these downward edges we obtain a path from v to the base vertex v_0 . This is an example of an **edgepath**, which is a composition of finitely many paths each consisting of a single edge traversed monotonically. For any edgepath joining v to v_0 other than the downward edgepath, the height function would not be monotone and hence would have local maxima, occurring when the edgepath backtracked, retracing some edge it had just crossed. Thus in a tree there is a unique nonbacktracking edgepath joining any two points. All the vertices and edges along this edgepath are distinct.

A tree can contain no subgraph homeomorphic to a circle, since two vertices in such a subgraph could be joined by more than one nonbacktracking edgepath. Conversely, if a connected graph X contains no circle subgraph, then it must be a tree. For if T is a maximal tree in X that is not equal to X , then the union of an edge of $X - T$ with the nonbacktracking edgepath in T joining the endpoints of this edge is a circle subgraph of X . So if there are no circle subgraphs of X , we must have $X = T$, a tree.

For an arbitrary connected graph X and a pair of vertices v_0 and v_1 in X there is a unique nonbacktracking edgepath in each homotopy class of paths from v_0 to v_1 . This can be seen by lifting to the universal cover $\tilde{X}$, which is a tree since it is simply-connected. Choosing a lift $\tilde{v}_0$ of v_0 , a homotopy class of paths from v_0 to v_1 lifts to a homotopy class of paths starting at $\tilde{v}_0$ and ending at a unique lift $\tilde{v}_1$ of v_1 . Then the unique nonbacktracking edgepath in $\tilde{X}$ from $\tilde{v}_0$ to $\tilde{v}_1$ projects to the desired nonbacktracking edgepath in X .

Exercises

1. Let X be a graph in which each vertex is an endpoint of only finitely many edges. Show that the weak topology on X is a metric topology.
2. Show that a connected graph retracts onto any connected subgraph.
3. For a finite graph X define the Euler characteristic $\chi(X)$ to be the number of vertices minus the number of edges. Show that $\chi(X) = 1$ if X is a tree, and that the rank (number of elements in a basis) of $\pi_1(X)$ is $1 - \chi(X)$ if X is connected.
4. If X is a finite graph and Y is a subgraph homeomorphic to S^1 and containing the basepoint x_0 , show that $\pi_1(X, x_0)$ has a basis in which one element is represented by the loop Y .
5. Construct a connected graph X and maps $f, g: X \rightarrow X$ such that $fg = \mathbb{1}$ but f and g do not induce isomorphisms on π_1 . [Note that $f_*g_* = \mathbb{1}$ implies that f_* is surjective and g_* is injective.]
6. Let F be the free group on two generators and let F' be its commutator subgroup. Find a set of free generators for F' by considering the covering space of the graph $S^1 \vee S^1$ corresponding to F' .